

Environmental Noise Assessment and Impact Analysis

**Department of Environmental Science and Engineering
Delhi Technological University, Delhi**

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1. Introduction

Environmental noise pollution is increasingly recognized as a critical environmental issue in urban and semi-urban regions, particularly in areas undergoing rapid infrastructural development.

Noise is defined as “any unwanted or harmful sound that disrupts normal human activities or degrades environmental quality”.

Sound itself is a “form of energy produced by vibrating bodies and transmitted through a medium such as air.” However, when this sound becomes excessive, irregular, or occurs at inappropriate times, it is classified as noise.

The human ear can perceive sound within a frequency range of approximately **20 Hz to 20,000 Hz**. The sensitivity of the human ear varies across this range, with maximum sensitivity typically observed between **250 Hz and 5000 Hz**. This variation plays an important role in how environmental noise is measured and evaluated.

1.1 Measurement of Noise and Key Concepts

Noise levels are quantified using the unit **decibel (dB)**, which is a logarithmic scale used to express sound intensity. This scale allows a wide range of sound levels to be represented in a compact form. However, the human ear does not respond equally to all frequencies; it is more sensitive to mid-frequency sounds and less sensitive to very low and very high frequencies.

To account for this variation in human hearing, different **frequency weightings** are used in noise measurements:

- **dB(A) (A-weighting):**
This is the most commonly used scale in environmental noise studies. It closely matches the sensitivity of the human ear by emphasizing mid-frequency sounds and reducing the contribution of very low and high frequencies. Hence, it is widely used for assessing traffic noise, industrial noise, and community noise.
- **dB(B) (B-weighting):**
This scale provides moderate weighting between A and C scales. It was designed for medium sound levels but is rarely used in modern noise measurements.
- **dB(C) (C-weighting):**
This scale gives nearly equal importance to all frequencies (almost flat response). It is used for measuring high intensity sounds such as explosions, machinery noise, and peak noise levels.

Thus, while **dB** represents the physical measurement of sound, weighted scales like **dB(A)**, **dB(B)**, and **dB(C)** adjust this measurement to reflect better human hearing, with **dB(A)** being the most widely used standard in environmental noise assessment.

1.2 Different Terminologies in Context of Sound Measurements

Level	Sound Intensity Level	Sound Pressure Level	Sound Power Level
Symbol	SIL	SPL	SWL
Express as	$10 \log (I / I_0)$	$20 \log (P / P_0)$	$10 \log (W / W_0)$
Units	SIL in dB, I in W/m^2	SPL in dB, P in Pa	SWL in dB, W in watts
*Reference Value	$I_0 = 1 \times 10^{-12} \text{ W/m}^2$ SIL = 0 dB	$P_0 = 20 \times 10^{-6} \text{ Pa}$ SPL = 0 dB	$W_0 = 1 \times 10^{-12} \text{ W}$ SWL = 0 dB
Applications	Environmental noise and Traffic Studies (L_{eq} , L_{10} , etc.)	Energy flow in sound waves	Source strength (machine noise)

Sound Pressure Level (SPL) is the most important parameter in environmental noise studies because it represents the sound as perceived by the human ear and can be directly measured using sound level meters. Most environmental noise indicators such as L_{eq} , L_{10} , L_{50} , L_{90} , and L_{dn} are based on SPL, making it the standard for noise assessment.

The energy of a sound wave is related to sound pressure as:

$$I = P^2 / (\rho c)$$

where:

I = sound intensity (W/m^2)

P = sound pressure (Pa)

ρ = density of medium

c = speed of sound

This shows that **sound intensity (energy) is proportional to the square of sound pressure ($I \propto P^2$)**, which is why SPL uses **20 log**, while SIL uses **10 log**.

As sound pressure increases, the energy of the sound wave increases significantly, meaning even small increases in pressure lead to large increases in energy. This relationship explains why sound pressure is a reliable indicator of noise impact and why SPL is widely used in environmental studies.

1.3 Important Noise Descriptors

To effectively assess environmental noise, several statistical indicators are used:

Parameter	Description
Leq	The constant sound level that would produce the same energy as the fluctuating noise over a given time
Lmax	Maximum recorded noise level during the observation period
Lmin	Minimum recorded noise level during the observation period
Ln	Noise level exceeded for n% of the total time, ex. L10, L50, L90
L10	Noise level exceeded for 10% of the time (indicates peak noise)
L50	Median noise level (50% of time) represents the average/typical background noise level, The noise level that is exceeded for 50% of the total measurement time
L90	Background noise level (90% of time) L90 is the level exceeded for 90% of the time. For 90% of the time, the noise level is above this level.
Ldn	24-hour average noise level where night noise (10 PM–6 AM) is penalized by +10 dB

Leq, L10, L50, L90, Ldn, Ln are all **SPL-based noise descriptors**.

Leq is particularly important as it represents a constant sound level that would produce the same energy as the fluctuating noise over a given time period .

These descriptors provide a comprehensive understanding of both the intensity and variability of noise in a given environment, making them essential tools in environmental noise analysis.

2. Theoretical and Regulatory Framework for Acoustic Monitoring in Educational Institutions

The acoustic environment of an educational institution is a fundamental component of its environmental health and pedagogical efficacy. Noise pollution, which is characterized by unwanted sound that disrupts human activity and well-being, is a significant environmental stressor in urban centers such as New Delhi. At Delhi Technological University (DTU), the expansion of academic infrastructure, specifically the construction of Academic Block 1 (AB-1), will introduce a complex set of noise sources into a sensitive social landscape. As an educational institution, the campus is designated as a "**Silence Zone**" under the **Noise Pollution (Regulation and Control) Rules, 2000**, which mandates the most stringent noise standards to protect the cognitive and physiological health of students and faculty.

The primary unit of measurement will be the A-weighted decibel (dBA), which adjusts sound pressure levels to match the human ear's frequency response. Because of the variable nature of environmental noise, the Equivalent Continuous Sound Level (Leq) will be the standard metric used to represent the average acoustic energy over a specific period. The determination of Leq is mathematically defined as:

$$Leq = 10 \cdot \log_{10} \left(\frac{1}{n} \sum_{i=1}^n 10^{\frac{Li}{10}} \right)$$

This logarithmic average ensures that high-intensity noise events, even if short in duration, are adequately reflected in the final value, as sound energy doubles every 3 dB increase and perceived loudness doubles every 10 dB. The Central Pollution Control Board (CPCB) of India provides the regulatory benchmarks for these values across different land-use categories. Although Leq is conventionally evaluated over an 8-hour period, in this study it has been calculated for shorter intervals (15 minutes) due to time constraints, while still providing a representative assessment of noise conditions

In addition to Leq , statistical noise descriptors (L_n) are also used to characterize the variability of environmental noise. These include L_{10} , L_{50} , and L_{90} , which represent the sound levels exceeded for 10%, 50%, and 90% of the measurement period

2.1 National Ambient Noise Standards and Compliance Targets

To regulate environmental noise and protect public health, the Government of India has established permissible noise limits under the **Noise Pollution (Regulation and Control) Rules, 2000**. These standards categorize areas based on land use and prescribe acceptable noise levels for daytime and nighttime.

Area Code	Category of Area	Daytime Limit (Leq dBA)	Nighttime Limit (Leq dBA)
(A)	Industrial Area	75	70

(B)	Commercial Area	65	55
(C)	Residential Area	55	45
(D)	Silence Zone	50	40

(Daytime: 6 AM – 10 PM)

(Nighttime: 10 PM – 6 AM)

Delhi Technological University, being an educational institution, falls under the “**silence zone category**”, where stricter noise limits are imposed. This makes it particularly important to monitor and control noise levels in construction areas such as AB-1. Silence zones specifically encompass areas within 100 meters of the premises of schools, colleges, hospitals, and courts. Any deviation from these limits by more than 10 dB(A) empowers authorities to take legal action and enables citizens to file formal complaints. Given the active construction at AB-1, the university would be put at risk of environmental non-compliance

3. Empirical Data Analysis and Location-Wise Interpretation

The noise survey was conducted at five locations around the AB-1 construction site. The data has been recorded in specialized "Noise Level Proformas," capturing fluctuations at 15-second intervals over 60-minute sampling periods. This high-resolution temporal data allows for the identification of specific impulsive noise events and the calculation of representative **L_{max}** (maximum) and **L_{min}** (minimum) values as well as **L(Eq)** values.

3.1 Instruments Used :-

Instant sound level meter is used for the collection of data related to noise at an interval of 15 seconds per reading for 1 hour , The instrument details are given below

Instrument	Make	Model	Detection Limit
Sound Level Meter (SLM)	Cesva	SC260	Lower Limit: 30 dBA Upper Limit: 137 dBA Peak: 140 db



3.2 Sampling Site Breakdown

All collected data corresponds to "Daytime" measurements, as sampling occurred between 11:37 AM and 4:00 PM across the various dates. No nighttime data was provided for analysis due to administrative constraints hence, limiting the assessment to tentative daytime active construction and academic hours.

Location ID	Site Name	Date	Time	Period	Leq (dba)
LOC-01	AB1 Near DOD	06-04-2026	4:00 PM	1st Hour	58.71
LOC-02	AB1 Near Library	06-04-2026	2:50 PM	1st Hour	55.96
LOC-03	CPCB Site (DTU) - Set 1	27-03-2026	11:37 AM	1st Hour	58.58
LOC-04	CPCB Site (DTU) - Set 2	27-03-2026	12:50 PM	1st Hour	64.84
LOC-05	Dept. of Science Building	06-04-2026	3:00 PM	1st Hour	72.53



Fig. Site Photo AB 1, DTU

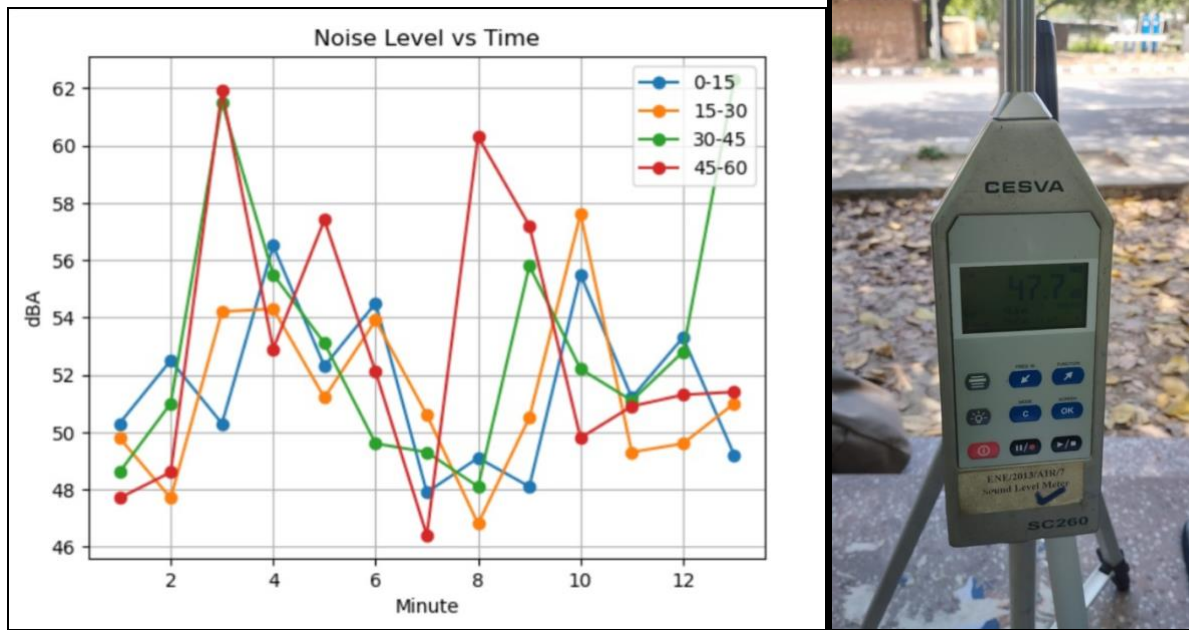
3.2.1 AB1 Near Department of Design (DOD)

The Department of Design is a sensitive academic receptor where creativity and precision work are central. The monitoring data reveals a highly fluctuating noise environment. The readings at LOC-01 frequently exceeded 50 dBA, with several peaks above 60 dBA and a maximum spike of 73.8 dBA in the final minutes of the hour.

The acoustic profile at DOD shows a

- **Lmin** value of 46.4 dBA

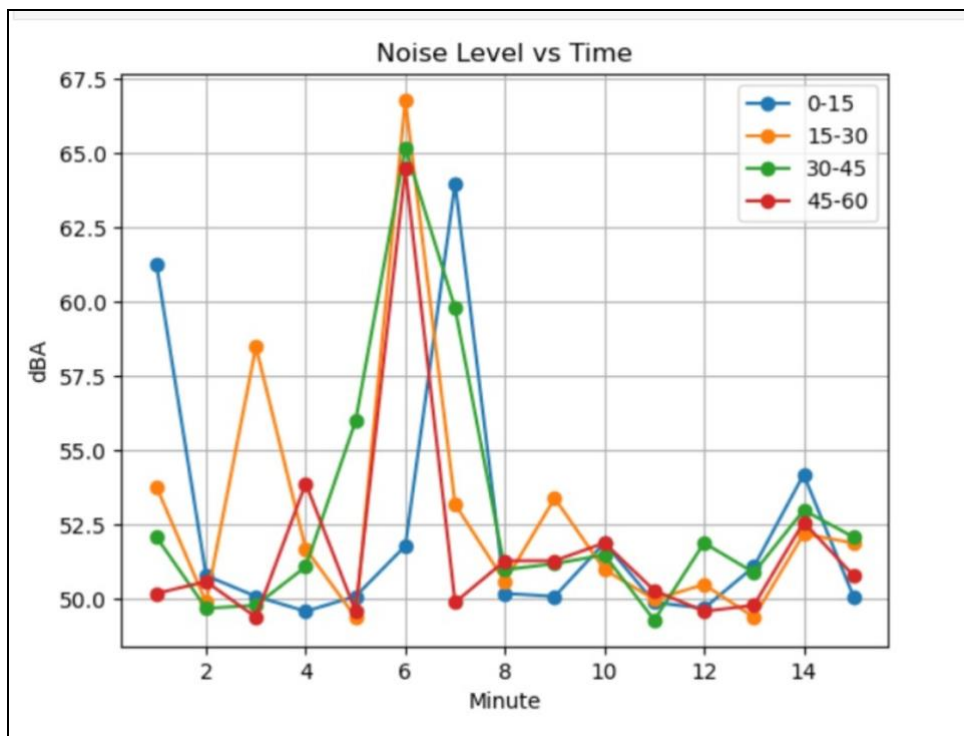
The majority of the 15-minute blocks recorded values in the range of 52 to 57 dBA. This indicates that the ambient level at DOD is already exceeding the CPCB silence zone limit by at least 2 to 7 dBA, even before accounting for the significant peaks that pushed the environment toward levels more typical of a commercial or industrial zone.



3.2.2 AB1 Near Central Library

The Central Library represents the most critical silence-dependent facility on campus. Data collected at LOC-02 provides a disturbing look at the acoustic stress faced by students:

- **Lmax:** 66.8 dBA .
- **Lmin:** 49.3 dBA



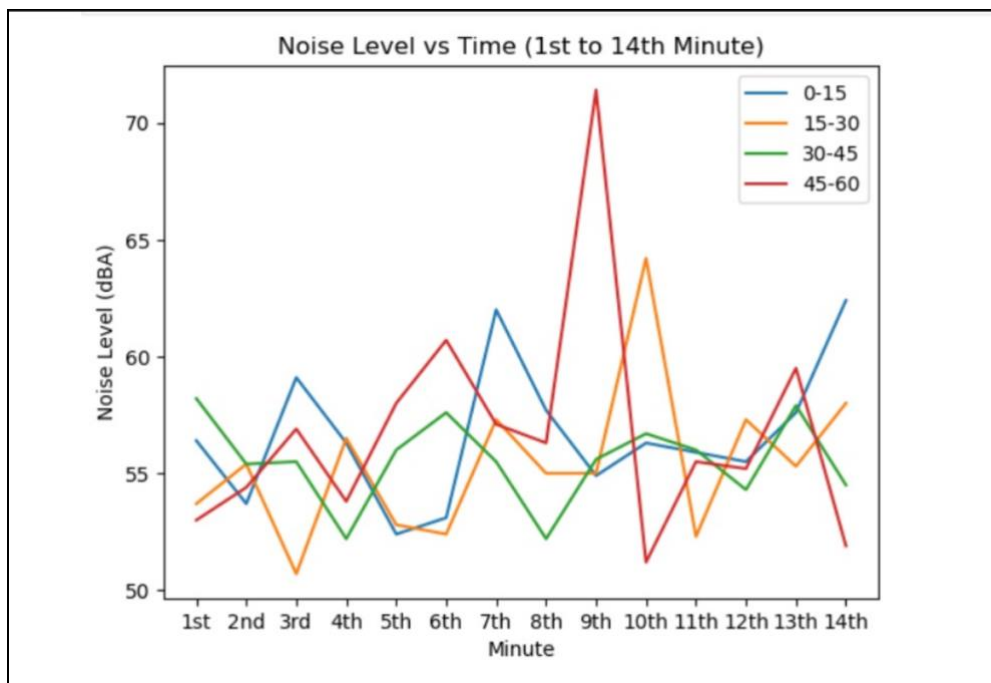


3.2.3 Site Analysis: CPCB Site (DTU) Reference Station Morning Session

The CPCB site data serves as a reference point for the campus core environment. This location was monitored twice during the peak midday period.

The morning session recorded an

- **L_{max}** of 71.4 dBA
- **L_{min}** of 50.7 dBA



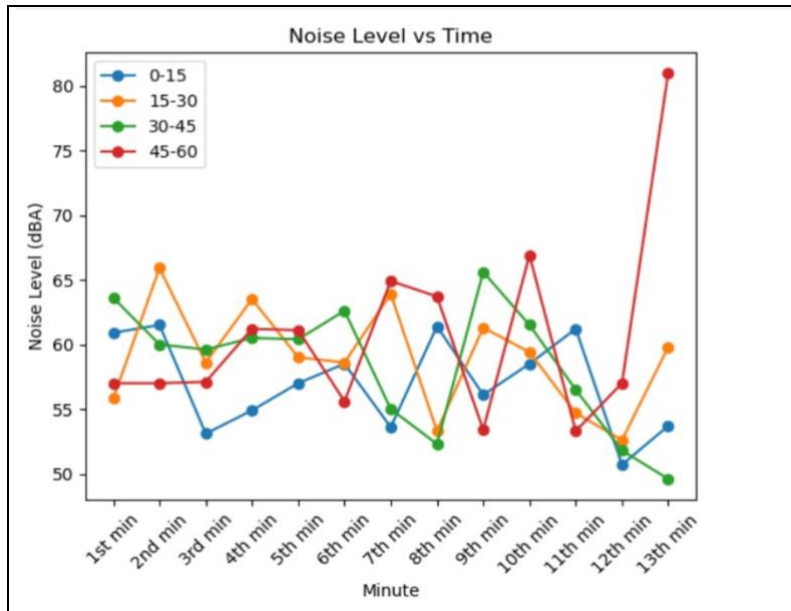


3.2.4 Site Analysis: CPCB Site (DTU) Reference Station Morning Session

The noise levels intensified during the second hour of monitoring.

- **L_{max}** of 81.0 dBA
- **L_{min}** of 49.6 dBA

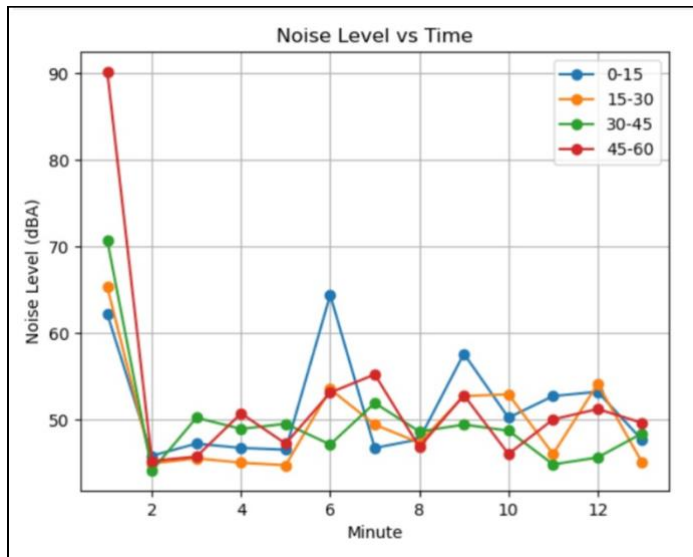
which is a critical violation of both academic and industrial standards. L_{min} indicates that while the base noise level is occasionally compliant, the environment is dominated by high-decibel events that raise the energy-average far above the 50 dBA target.



3.2.5 Site Analysis: Department of Science Building

The Department of Science Building monitoring session at LOC-05 yielded the most extreme data points in the entire study:

- **Lmax:** 90.2 dBA (recorded in the 1st minute of the final interval).
- **Lmin:** 44.1 dBA.



The 90.2 dBA reading is an outlier of extreme intensity, yet it is representative of a real-world occurrence at the site. While the **L_{min}** of 44.1 dBA was the lowest recorded in the study, the massive range (a delta of 46.1 dBA) suggests a highly erratic acoustic environment.

4. Noise Contour Mapping

This noise contour map represents the spatial distribution of equivalent noise levels (L_{eq}) across the AB-1 study site and surrounding areas at Delhi Technological University. The map was generated using interpolated sound measurements (Inverse Distance Weighted - IDW of a point vector layer) collected from selected sampling locations within the study area. Noise levels range from approximately 57 dB(A) to above 71 dB(A), with higher noise intensity observed near the Department of Science Building and relatively lower levels around the library and Department of Design.

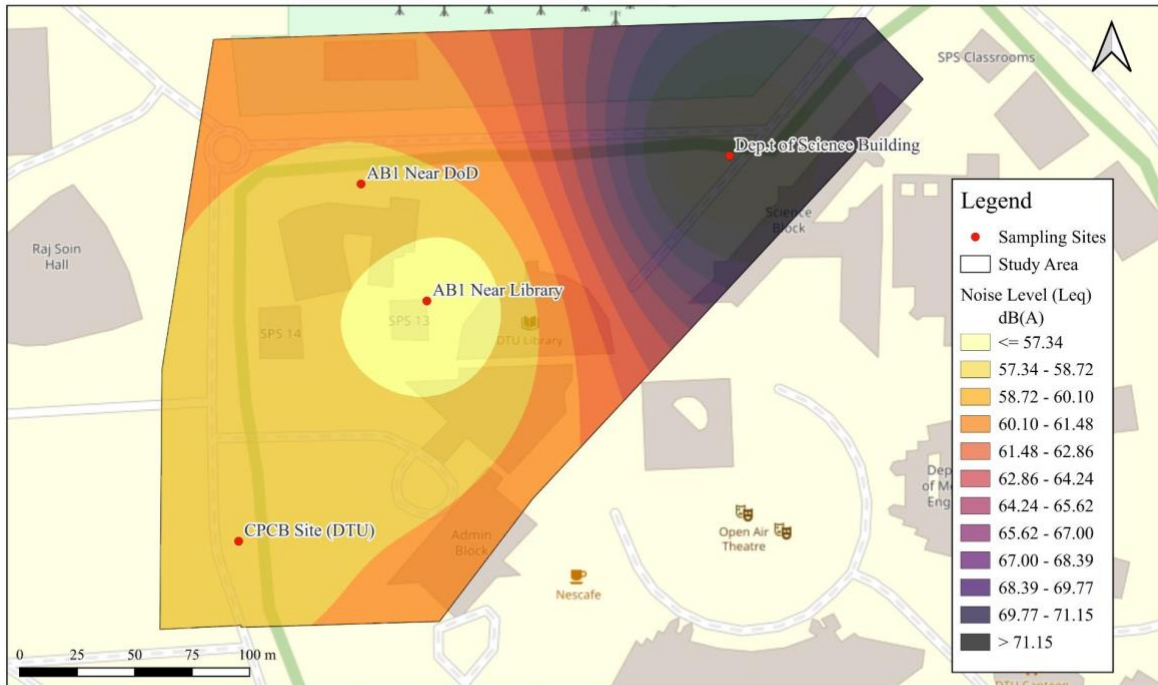
Methodology for Noise Contour Mapping in QGIS is as follows.

- A new project was first created in QGIS to establish the workspace for the noise contour mapping process.
- Subsequently, a new GeoPackage point layer (vector data type) was created to represent the sampling locations. The point features were then carefully digitized according to the coordinates of the noise monitoring sites. Relevant attribute information, including the recorded noise level values, was entered into the attribute table for each sampling point.
- After preparing the point dataset, the Inverse Distance Weighted (IDW) interpolation method was selected from the Processing Toolbox. This interpolation technique was used to estimate the spatial distribution of noise levels across the study area based on observations from the four sampling sites. The Leq (Equivalent Continuous Sound Level) field was used as the input parameter for interpolation.
- As a result, an interpolated raster layer representing the continuous spatial variation of noise levels was generated.
- Next, a new GeoPackage polygon layer (vector data type) was created to delineate the boundary of the study area. The study area extent was then digitized accurately as a polygon feature.
- The interpolated raster layer was subsequently clipped using the study area polygon, ensuring that the final output was restricted only to the area of interest.
- For visualization purposes, appropriate layer styling and symbology were applied to the output raster layer. The Equal Interval classification method was selected for contour representation, with the number of classes set to 12. The Inferno color ramp was chosen to effectively display the variation in noise intensity levels.
- Finally, a map layout was created to prepare the output for presentation and reporting. Essential cartographic elements such as the legend, scale bar, north arrow, date, and map credits were added to produce a clear and professional map output.

	fid	location_id	site_name	Leq	Lmin	Lmax	time
1	1	LOC-01	AB1 Near DoD	58.71	46.4	73.8	4/6/2026 16:00:00 ...
2	2	LOC-02	AB1 Near Library	55.96	49.3	66.8	4/6/2026 14:50:00 ...
3	3	LOC-03	CPCB Site (DTU)	58.58	50.7	71.4	3/27/2026 11:37:00 ...
4	4	LOC-05	Dep.t of Science Building	72.53	44.1	90.2	4/6/2026 15:00:00 ...

Fig. Attribute Table of Sampling Sites

Noise Contour Map of the AB-1 Site in Delhi Technological University



Prepared by: EIA Survey Camp Group 4 (2025-2026),
Department of Environmental Science and Engineering,
Delhi Technological University.
Date: 14/04/2026

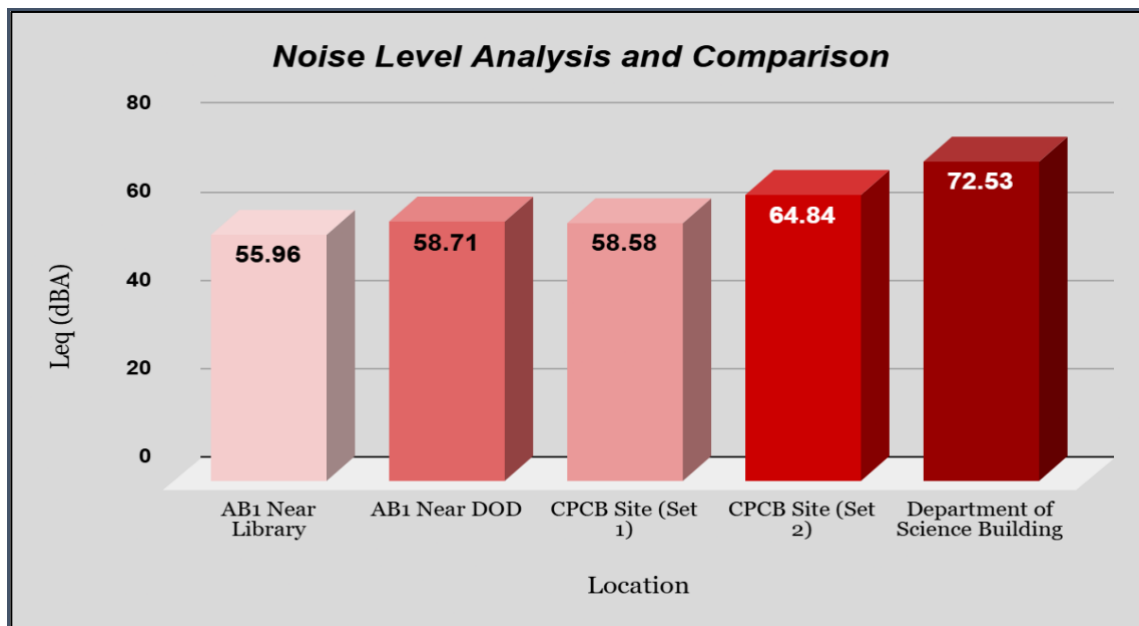
Fig. Noise Contour Map

5. Comparative Analysis and Compliance Status

Monitoring Location	L10	L90	Observed Lmax (dBA)	Observed Lmin (dBA)	Observed Leq	CPCB Limit (Day)	Compliance Status	Deviation (Max)
AB1 Near DOD	60.30	48.10	73.8	46.4	58.71	50	Non-Compliant	+23.8 dBA

AB1 Near Library	59.15	49.60	66.8	49.3	55.96	50	Non-Compliant	+16.8 dBA
CPCB Site (Set 1)	60.58	52.31	71.4	50.7	58.58	50	Non-Compliant	+21.4 dBA
CPCB Site (Set 2)	63.88	52.41	81.0	49.6	64.84	50	Non-Compliant	+31.0 dBA
Dept. of Science	60.56	45.02	90.2	44.1	72.53	50	Non-Compliant	+40.2 dBA

The observed noise data shows a breach of the CPCB Ambient Noise Standards for Silence Zones. The following table showcases a direct comparison of the recorded parameters across all five locations against the national regulatory limit of 50 dBA for daytime academic environments.



6. Regulatory Compliance Status

Therefore environmental noise assessment of the AB-1 construction zone at Delhi Technological University yields the following findings:

- **Non-Compliance:** The site is in absolute non-compliance with the CPCB Ambient Air Quality Standards in Respect of Noise for Silence Zones. Daytime levels consistently exceed the 50 dBA limit by margins ranging from 16 to 40 dBA.
- **Hazardous Peaks:** The maximum recorded value of 90.2 dBA at the Department of Science Building represents a significant environmental and occupational hazard.
- **Inadequate Mitigation:** The high intensity of peaks recorded at the Library and DOD locations suggests that existing noise barriers or operational restrictions are either non-existent or fundamentally ineffective.
- **Legal Risk:** Under the current enforcement climate in Delhi, DTU and its contractors face significant legal and financial liability if corrective measures are not immediately implemented, as regulatory authorities have historically imposed heavy fines on construction sites for dust and noise violations.

7. Technical Interpretation

7.1 Critical Hotspots and Violation Assessment

The comparison analysis reveals that **all of the monitored locations** are in violation of the CPCB's **50 dBA silence zone daytime limit** during active hours.

The Department of Science Building is identified as the most critical hotspot, showing a deviation of more than 40 dBA from the permissible limit.

Another observation is that at several locations, including the Library and DOD, the *minimum* recorded noise level was frequently near or above the *maximum* permissible average, indicating that there is no effective restorative quiet period for the receptors.

The CPCB site data suggests a temporal trend where noise levels escalate significantly between late morning and early afternoon (from 71.4 dBA to 81.0 dBA), likely coinciding with increased sound due to passing traffic and students.

As per the Noise Pollution Rules, 2000, any violation exceeding 10 dBA above the ambient standard is considered a major breach that requires immediate intervention by the designated authority.

7.2 Existing Sources of Noise in Construction Lifecycle (Baseline Conditions)

Before the initiation of any construction activity, the study area already has certain background noise levels known as **baseline noise**. These sources must be considered to understand the **incremental impact of construction**.

- **Traffic Noise**
Movement of cars, bikes, DTU Golf Cart within campus.
- **Academic and Institutional Activities**
Classroom discussions, student gatherings, announcements.
- **Mechanical Equipment in Buildings**
Air conditioners, generators, water pumps.

- **Human Activities**
Conversations, walking, social interactions.
- **Nearby Urban Influence**
Noise from surrounding roads, markets, or residential areas.
- **Natural Sources**
Wind, birds, and other environmental sounds.

7.3 Impact of Noise

Excessive exposure to noise has several adverse effects on human health, performance, and the environment. The following major impacts:

Impact on Research and Laboratory Work

- The Department of Science utilizes precision equipment that can be sensitive to mechanical vibrations and the high sound pressure levels that can interfere with micro-measurements
- Academic performance is directly linked to the acoustic environment. Research indicates that noise levels above 55 dBA adversely affect reading comprehension and information processing

Physiological and Psychological Effects

- Annoyance and irritation
- increase stress hormone levels and anxiety
- hypertension and cardiovascular issues
- Headaches and fatigue
- Sleep disturbances
- Reduced concentration and work efficiency

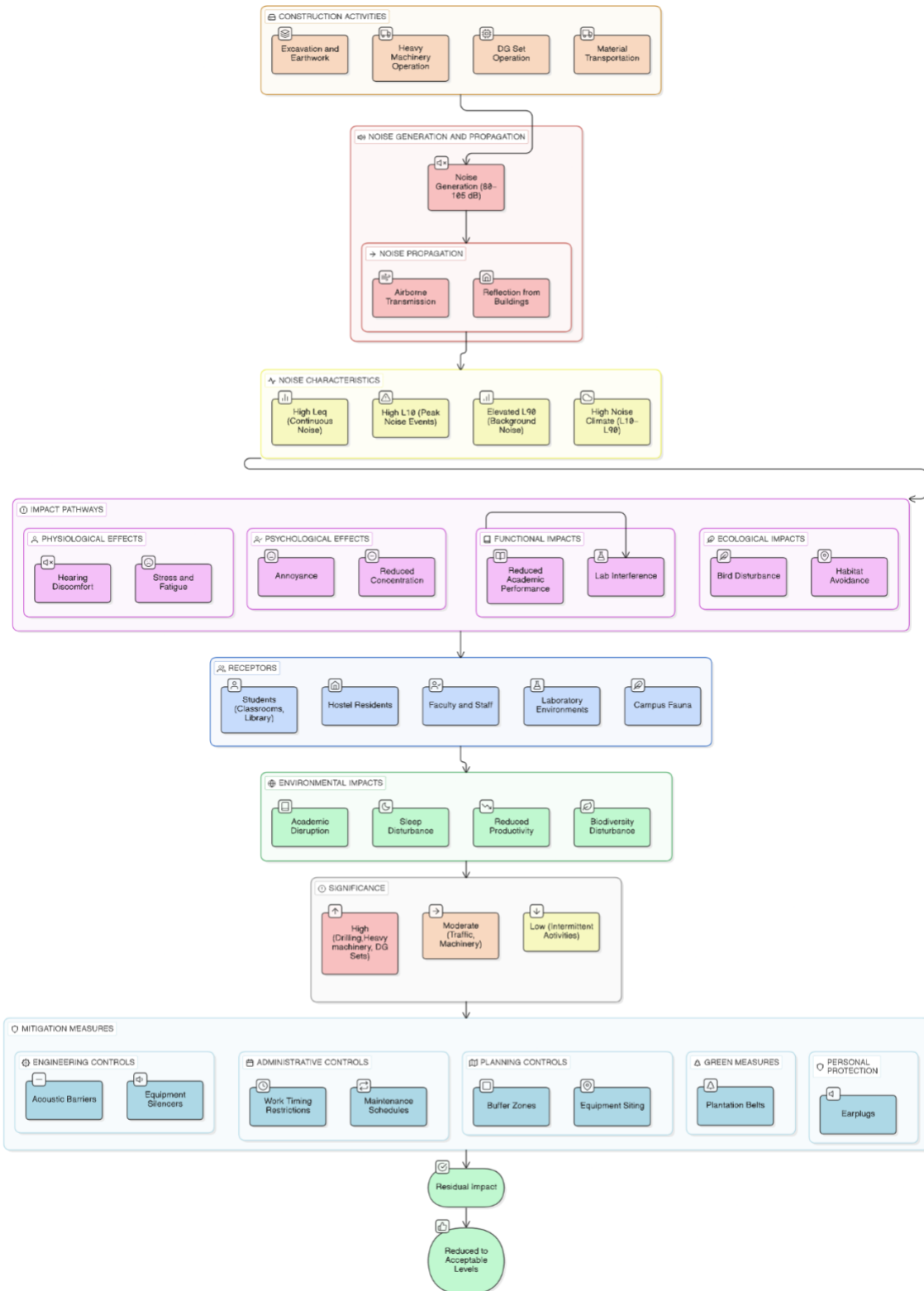
Auditory Effects

- Temporary or permanent hearing loss
- Increased sensitivity to loud sounds

Environmental and Social Effects

- Disturbance to surrounding population
- Interference with communication
- Reduced academic productivity in educational institutions

Impact Network Diagram



8. Noise Mitigation and Environmental Management

To achieve compliance and protect sensitive receptors, the following technical and administrative measures must be implemented.

8.1 Engineering and Technical Solutions

1. **High-Performance Noise Barriers:** Temporary acoustic barriers should be installed around the entire perimeter of the AB-1 construction site. These barriers should consist of mass-loaded vinyl or specialized acoustic panels rather than standard plywood. Barriers should be at least 3–4 meters high to break the line-of-sight between machinery and the windows of nearby academic buildings. Properly designed barriers can reduce noise by 10–15 dBA.
2. **Acoustic Enclosures for Stationary Equipment:** All generators, compressors, and pumps must be housed in soundproof acoustic enclosures. For older machinery, high-performance exhaust mufflers with a minimum insertion loss of 25 dBA should be retrofitted.
3. **Use of "Quiet" Technology:** The contractor should be mandated to use modern, low-noise equipment. For example, electric concrete vibrators with vibration-damping handles should be utilized to reduce noise at the source.
4. **Acoustic Treatment of Receptors:** For critical areas like the Central Library and Science Building labs, the university should consider secondary glazing or sealing of windows to improve the building's sound attenuation factor.

8.2 Administrative and Operational Controls

1. **Strategic Scheduling of High-Noise Tasks:** Noisy activities (drilling or demolition) must be prohibited during peak academic hours and completely banned during university examination periods. Scheduling these tasks for early morning or late afternoon (within daytime limits) can reduce the impact on students.
2. **Equipment Positioning and Orientation:** Loudest machines should be positioned as far as possible from the Library and hostel blocks. Engines and exhaust outlets should be oriented away from academic buildings to utilize natural distance attenuation.
3. **Mandatory PPE for Workers:** All workers operating in zones where noise exceeds 85 dBA must use hearing protection (earplugs or earmuffs) to prevent occupational hearing loss.
4. **Real-Time Monitoring and Transparency:** Install permanent, real-time noise monitors at the Library and Science Building. These should be linked to a public display or a web portal to ensure transparency and accountability.

Implementation of these measures will bring the AB-1 project into closer alignment with national environmental laws and protect the well-being of the DTU community.

9. Generation of Checklist for Noise and Vibration, and Emissions of Light and Heat

(As per EIA Notification, 14th September 2006 – Form I for EC)

S. No.	Information/Checklist Confirmation	Yes/No	Details thereof (with approximate quantities/rates, wherever possible) with source of information												
1	From the operation of equipment (engines, ventilation plant, generators, etc.)	Yes	Noise will be generated from construction machinery such as excavators, concrete mixers, DG sets, and material handling equipment.												
2	From industrial or similar processes	No	Not Applicable (Project is construction-based within academic campus).												
3	From construction or demolition	Yes	<table><tr><th>Parameter</th><th>Observed Range (dBA)</th></tr><tr><td>Lmin</td><td>44.1 – 50.7</td></tr><tr><td>L90</td><td>45.02 – 52.41</td></tr><tr><td>L10</td><td>59.15 – 63.88</td></tr><tr><td>Leq</td><td>55.96 – 72.53</td></tr><tr><td>Lmax</td><td>66.8 – 90.2</td></tr></table>	Parameter	Observed Range (dBA)	Lmin	44.1 – 50.7	L90	45.02 – 52.41	L10	59.15 – 63.88	Leq	55.96 – 72.53	Lmax	66.8 – 90.2
Parameter	Observed Range (dBA)														
Lmin	44.1 – 50.7														
L90	45.02 – 52.41														
L10	59.15 – 63.88														
Leq	55.96 – 72.53														
Lmax	66.8 – 90.2														
4	From blasting or piling	No	No blasting activities observed during the monitoring period.												
5	From construction or operational traffic	Yes	Noise from movement of trucks, dumpers, and campus vehicles (approx. 70–80 dBA) contributes significantly to ambient noise levels, especially during peak hours.												

6	From lighting or cooling systems	No	Not significant before the construction phase.
7	From any other sources	Yes	Background noise from academic activities, human movement, and nearby urban traffic contributes to baseline noise levels.

10. Conclusion

The Environmental Noise Assessment of the AB-1 construction site at Delhi Technological University clearly demonstrates that the study area is operating under **significant acoustic stress and persistent regulatory non-compliance** with the CPCB-prescribed silence zone limit of 50 dBA for daytime conditions.

Analysis of monitored data indicates that **all five locations consistently exceeded permissible limits**, with Leq values ranging from approximately 55 dBA to 72 dBA, and extreme peak levels reaching **up to 90.2 dBA**. These findings confirm that the noise environment is not only elevated but also **highly dynamic and dominated by impulsive and intermittent construction activities**.

Particularly concerning is the exposure of **sensitive receptors**, including the Central Library, Department of Design, and Department of Science Building. The absence of any sustained low-noise periods (as reflected by Lmin values approaching or exceeding regulatory thresholds) suggests a **continuous background disturbance**, which can significantly impair cognitive performance, concentration, and research productivity in an academic setting.

From a health and environmental perspective, prolonged exposure to such elevated noise levels may lead to **physiological and psychological impacts**, including stress, fatigue, reduced efficiency, and potential hearing-related risks. Furthermore, peak noise events above 85–90 dBA pose **occupational hazards for on-site workers**, necessitating immediate protective interventions.

The study also highlights a critical gap in **existing mitigation and management practices**, as current construction activities lack adequate noise control measures such as engineered barriers, acoustic enclosures, and optimized equipment usage. This not only intensifies environmental degradation but also increases the risk of **legal and regulatory action** under the Noise Pollution (Regulation and Control) Rules, 2000, where exceedances beyond 10 dBA are treated as serious violations.

To address these concerns, it is essential to implement a comprehensive **Noise Management and Mitigation Framework**, integrating:

- **Engineering Controls:** Installation of high-performance acoustic barriers, enclosure of noise-generating equipment, and adoption of low-noise technologies
- **Administrative Measures:** Strategic scheduling of high-noise activities, zoning of equipment, and restriction during peak academic hours
- **Monitoring Mechanisms:** Deployment of real-time noise monitoring systems to ensure transparency and compliance



APPENDIX

A. DETAILED METHODOLOGY FOR CALCULATION OF NOISE INDICES

A.1 Introduction

This annexure presents the **detailed methodology adopted for the calculation of statistical noise indicators**, including Equivalent Continuous Sound Level (Leq), L10, and L90, based on field observations conducted at the AB-1 construction site. The procedure follows standard acoustic principles and CPCB-recommended practices for environmental noise assessment.

A.2 Data Collection Framework

Noise monitoring was carried out using a calibrated **Sound Level Meter (SLM)** at selected locations. Measurements were recorded at **15-second intervals over a duration of 60 minutes**, resulting in a total of:

N= 240 observations per monitoring location

The collected dataset forms the basis for all subsequent calculations.

A.3 Calculation of Equivalent Continuous Sound Level (Leq)

The Equivalent Continuous Sound Level (Leq) represents the **constant sound level that contains the same acoustic energy as the varying noise over a given period**.

$$L_{eq} = 10 \log_{10} \left(\frac{1}{N} \sum_{i=1}^N 10^{\frac{L_i}{10}} \right)$$

The Leq is computed using the logarithmic expression:

Where:

- Li = Individual observed sound level (dBA)
- N = Total number of observations

This formulation ensures that **higher noise levels contribute disproportionately to the overall average**, reflecting their greater impact.

A.4 Determination of Statistical Noise Levels (L10 and L90)

Statistical noise levels provide insight into the **distribution and variability of noise over time**.

- **L10:** Sound level exceeded for 10% of the total observation period (indicator of peak noise conditions)
- **L90:** Sound level exceeded for 90% of the total observation period (indicator of background noise)

A.5 Calculation Procedure for L10 and L90

1. All recorded noise levels are arranged in **descending order of magnitude**.

$$\text{Position} = \frac{n}{100} \times N$$

2. The positional value is determined using:

Where:

- $N = 10$ for L10 and $n = 90$ for L90
- $N = 240$

3. Accordingly:

L10 = 24th highest value

L90 = 216th value

A.6 Interpretation of Results

- A **higher L10 value** reflects the presence of intermittent high-intensity noise sources such as construction equipment and vehicular movement.
- A **higher L90 value** indicates elevated background noise levels and lack of acoustic recovery periods.
- A significant difference between L10 and L90 suggests a **highly fluctuating noise environment**, typical of active construction zones.

A.7 Relevance to Present Study

The computed values of L10 and L90 for all monitoring locations indicate that the study area is subjected to **both high peak noise events and elevated baseline noise levels**, confirming persistent acoustic disturbance.

B. References

- **MoEFCC (2006)**
Environmental Impact Assessment Notification, 14th September 2006
Ministry of Environment, Forest and Climate Change, Government of India
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Central Pollution Control Board, India
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Guidelines for Ambient Noise Monitoring
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Acoustics — Description, Measurement and Assessment of Environmental Noise
- **Form 1 (EIA Notification 2006)**